Agreement depends on lexical access in at-a-glance reading: An EEG study in Spanish

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Introduction. Typically, language comprehension is characterized as proceeding 'word-by-word' [1-2]. However, behavioral and MEG/EEG recordings show that readers distinguish short sentences vs. non-sentences viewed 'at-a-glance' (200–300ms) [3-7]. Neural responses often are reported during the 'N400' time window, corresponding to lexical activation [8]. This has been taken as evidence for rapid, parallel lexical access that simultaneously facilitates a syntactic representation [3-4], though other results suggest different conclusions with respect to the relation between syntactic structure building vs. lexical access [5-7]. Here, we report on an 'at-a-glance' EEG study in Spanish that uses two-word noun-adjective combinations that have a gender agreement dependency (*coche*$_{[\text{MASC}]}$ *rápid-**o***$_{[\text{MASC}]}$ 'fast car'), which allows for us to probe the relation between syntactic and morphological/lexical processing. We find that the brain's sensitivity to agreement coincides with sensitivity to lexical surprisal and orthographic neighborhood density, (~300ms), but precedes sensitivity to syntactic structure (~600ms). We take this to demonstrate that, <u>in at-a-glance reading, some morphosyntactic computations can be executed in parallel with lexical access</u>.

Methods. Participants (*N* = 18/30 planned) read two-word phrases in Spanish, each displayed for 300ms (800ms) ISI (Fig 1A). Each phrase was followed by another, and participants judged whether they matched (50% match) [5]. There were 132 sets of phrases, in 3 conditions (Fig 1C): grammatical noun-adjective pairs, ungrammatical noun-adjective pairs with gender agreement violations, and ungrammatical noun-noun lists with no licit syntactic structure (Table 1). EEG activity was continuously recorded using MagStim EGI 256ch EEG.

Analysis. Epochs (–100-800ms) were averaged during presentation of each two-word phrase. Raw EEG activity was fit to a linear regression at each time point and sensor across items within participants (Fig 1B) during the N400 time window (200–500ms), yielding a beta coefficient for each regressor. *T*-tests were conducted at each time point and sensor across participants to determine whether they were significantly different than zero. Test statistics (*t*) of significant points that were adjacent in space-time were clustered together. Spatio-temporal cluster-based permutation tests were conducted to correct for multiple comparisons for cluster size [9]. Analyses for lexical variables were conducted only on grammatical trials, analyses probing for neighborhood effects and agreement were conducted on grammatical and ungrammatical trials, and analyses for syntactic structure building were conducted on a subset of grammatical trials that were left out of the other analyses ('localizer' trials).

Results. Significant clusters for the surprisal of word 2 ($-\log_2 p(w_2 \mid w_1)$) and 'backwards' surprisal of word 1 ($-\log_2 p(w_1 \mid w_2)$) were observed in left anterior and left posterior sensors respectively (ps < 0.05, Fig 1D(i)-(ii)). Detection of ungrammatical agreement was observed in a marginally significant cluster in overlapping posterior midline sensors (p = 0.07, Fig 1D(iii)). The proportion of the orthographic neighborhood of word 1 matching in gender ($c(\text{neighbors}_{[\text{gender=W1gender}]})$ / $c(\text{neighbors})$) also yielded a cluster in this region in time and space (p = 0.06, Fig 1D(iv)). Pairwise *t*-tests over averaged EEG data in these clusters show that ungrammatical noun-adjective trials exerted greater positive activity than grammatical noun-adjective trials (ps < 0.05), but grammatical noun-adjective trials did not differ from ungrammatical noun-noun trials (ps > 0.05). No clusters for syntactic structure (noun-adjective phrases vs. noun-noun phrases) were observed in the N400 time window; a *post-hoc* analysis identified a cluster later (~600–800ms), in anterior midline sensors (p = 0.04, Fig 1D(v)).

Conclusion. In 'at-a-glance' reading in Spanish, readers can access two words in parallel. Moreover, sensitivity to morphosyntactic violations and morphosyntactic neighborhood density effects were distributed in a similar time/spatial region as these surprisal effects. These were distinct from the contrast between noun-adjective and noun-noun lists. We propose that (some) local morphosyntactic dependencies may be 'bootstrapped' from mechanisms underpinning lexical access, which may be distinct from and precede constituent structure building.

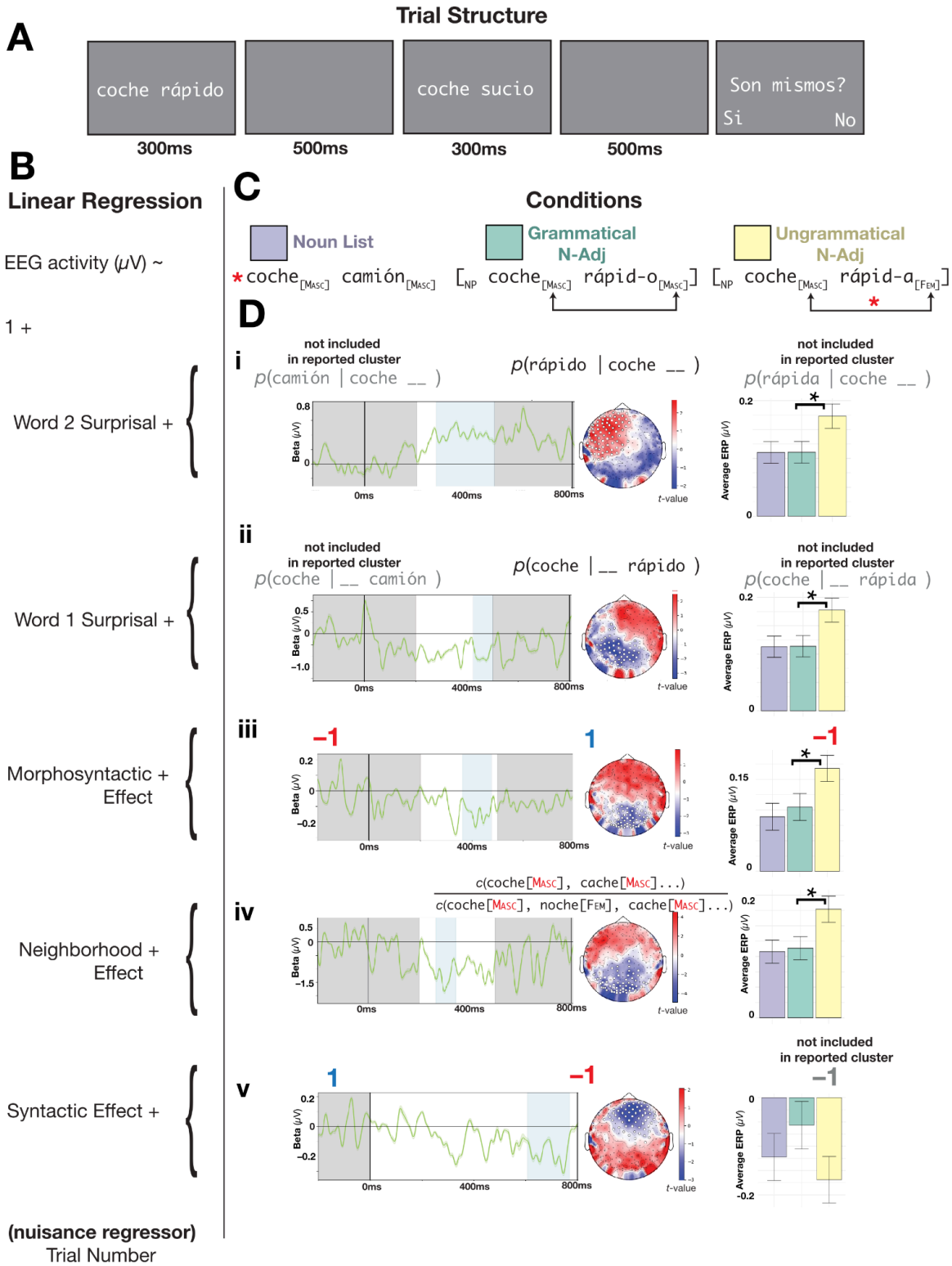

Fig 1: A: Task structure. **B:** Linear Regression Model**. C:** Experimental Design. **D:** Results. Time-courses show betas of raw EEG activity. Areas in white are time-windows used for clustering. Light blue areas are significant time points. Topoplots show the distribution of *t*-values on the sensor space. Bar plots show the average EEG activity in the cluster. **(i)** W2 Surprisal. **(ii)** W1 Surprisal **(iii)** Detection of agreement error. **(iv)** Effects of orthographic neighborhood with same gender features. **(v)** Detection of constituent structure.

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